

SUMMARY

- Backend & platform software engineer, 7+ years. Builds high-performance backend systems — APIs, data pipelines, ML-powered services — and the auth, observability, and data foundations teams build on. M.Sc. Computer Science, Politecnico di Milano (110 cum laude).

EXPERIENCE

- Duferco** Lugano, Switzerland
Software Engineer 03/2025 - Present
 - Defined the **company-wide backend standards** (auth, observability, data modeling) and shipped them as a shared **Python** platform library adopted across services.
 - Cut a core **scheduling/optimization batch** from **30 to 2 minutes (15x)** by porting the hot path to a native **Rust** extension, cutting the heaviest query latency **96%**, and parallelizing I/O.
 - Built the service backend (async **FastAPI** + **AWS** batch pipeline) with a concurrency-safe locking barrier and **MILP** optimization (Gurobi), covered by **700+ automated tests**.
 - Built a **Java/Spring Boot** middleware exposing a legacy enterprise ERP over HTTP (JSON + **Apache Arrow** streaming), removing client-side DB drivers across services.
 - Built an **LLM** document-extraction service (**Claude** on AWS Bedrock) converting business documents into validated structured data.
- Jobtome** Stabio, Switzerland
Software Engineer 11/2021 - 02/2025
 - Built **search APIs** in **Python** over **Elasticsearch**, sustaining **1,000 requests/second**.
 - Architected a **Python** job-crawler platform on **GCP** — **5,000+ crawlers**, **300k+ jobs/day** — plus **Apache Beam** pipelines ingesting **millions of records/day** (Pub/Sub to BigQuery).
 - Built and deployed an **NLP** classifier (**BERT**, **CNN**) on **Vertex AI**, processing **about 1M jobs/day** at **95% accuracy**.
- Polytechnic University of Milan** Milan, Italy
Research Fellow 10/2019 - 10/2021
 - Built **NLP** data pipelines analyzing **3M+ Italian social-media conversations** on vaccine discourse, and insights from reviews across **100 museums**.

SKILLS

- Languages:** Python, Rust, Java, SQL
Backend: FastAPI, Pydantic, asyncpg, Spring Boot, PostgreSQL, REST, OAuth2/JWT
Data & ML: Pandas, NumPy, PyTorch, BERT/Transformers, LLMs (Claude, AWS Bedrock), Apache Beam, BigQuery
Cloud & Infra: AWS, GCP, Docker, Kubernetes, OpenTelemetry, CI/CD, distributed systems

EDUCATION

- Polytechnic University of Milan** Milan, Italy
M.Sc. in Computer Science and Engineering; Final grade: 110 with honors/110 09/2016 – 04/2019
- Polytechnic University of Milan** Milan, Italy
Bachelor Engineering of Computing Systems 09/2013 – 09/2016

SELECTED PUBLICATIONS

- Pavanetto, Brambilla:** *Generating Realistic Web-Navigation Paths with RNNs and GANs*. ICWE 2020.
- Di Giovanni, Pavanetto, et al.:** *Content-based Classification of Vaccine-related Stances on Twitter (Italian)*. ICWSM 2021.